Support Vector Machines

Task: Binary classification

Consider a sample — & loss corresponding loss

$$\mathcal{L}' = \text{loss} = -y_i \log \hat{y}_i - (1-y_i) \log(1-\hat{y}_i)$$

where y_i = true class label of ith sample Class-1 $y_i=1$, Class-0 $y_i=0$

$\hat{y}_i$ = predicted probability

$$= \text{Prob}\left(y_i = 1 / \underline{x}_i, \underline{\theta}, b\right) = \frac{1}{1+e^{-(\underline{x}_i^T \underline{\theta} + b)}}$$

$i = 1, 2 \ldots n$

$$\underline{x}_i = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}_{p \times 1} \text{ feature vector} \qquad \underline{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

$$b + \underline{x}_i^T \underline{\theta} = \langle \underline{x}_i, \underline{\theta} \rangle + b = b + x_{i1}\theta_1 + x_{i2}\theta_2 + \ldots + x_{ip}\theta_p$$

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$$\mathcal{L}' = -y_i \log \hat{y}_i - (1-y_i) \log (1-\hat{y}_i)$$

$$= \begin{cases} -\log \hat{y}_i & \text{if } y_i = 1 \leftarrow \\ -\log (1-\hat{y}_i) & \text{if } y_i = 0 \end{cases}$$

$\hat{y}_i \in [0,1]$

Given Sample $y_i = 1$

$\hat{y}_i \longrightarrow 1$

$-\log 1 = 0$

$\mathcal{L}' \longrightarrow 0$

$\mathcal{L}' \longrightarrow \infty$

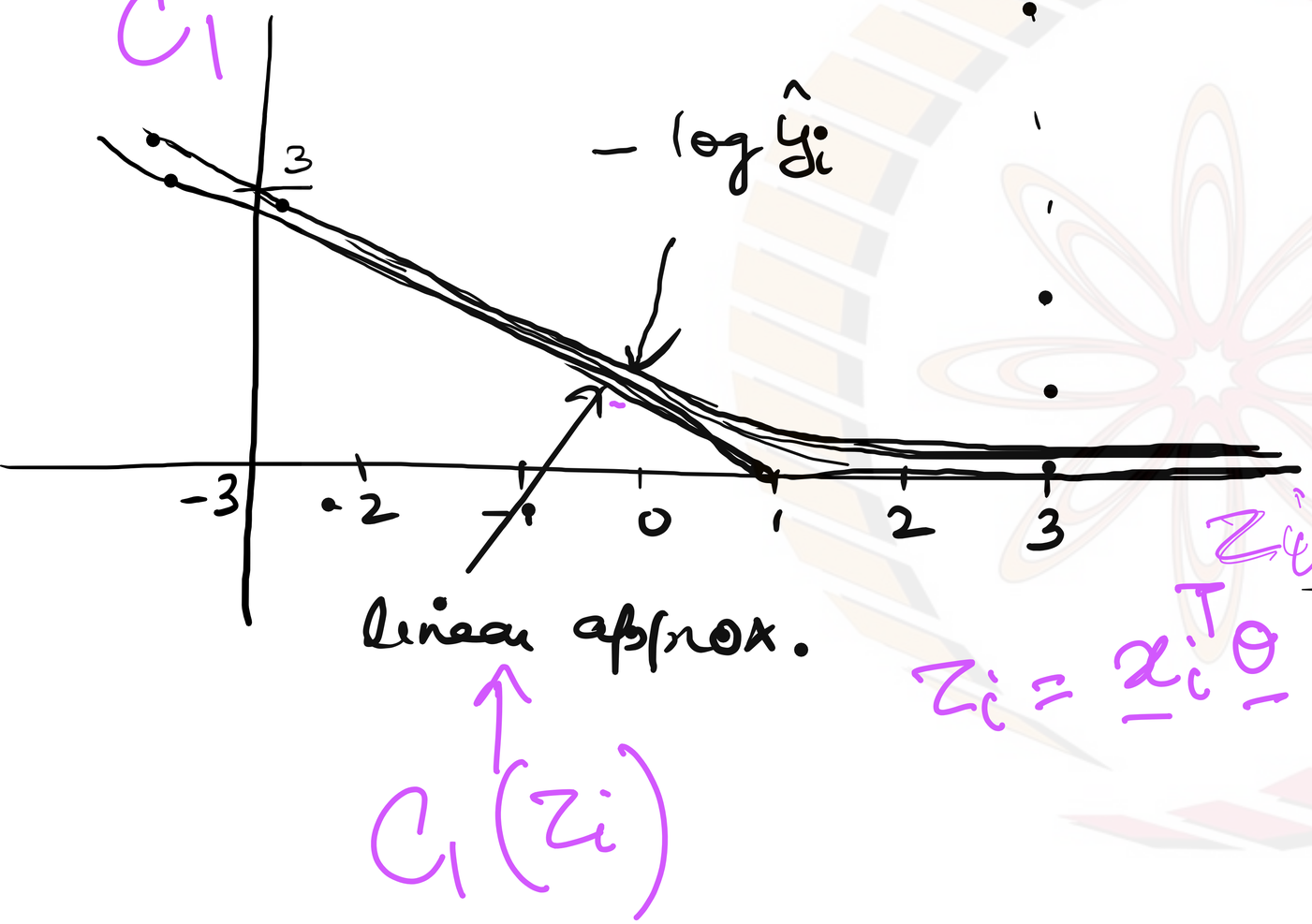
if $\hat{y}_i \to 1$ & $y_i = 1$

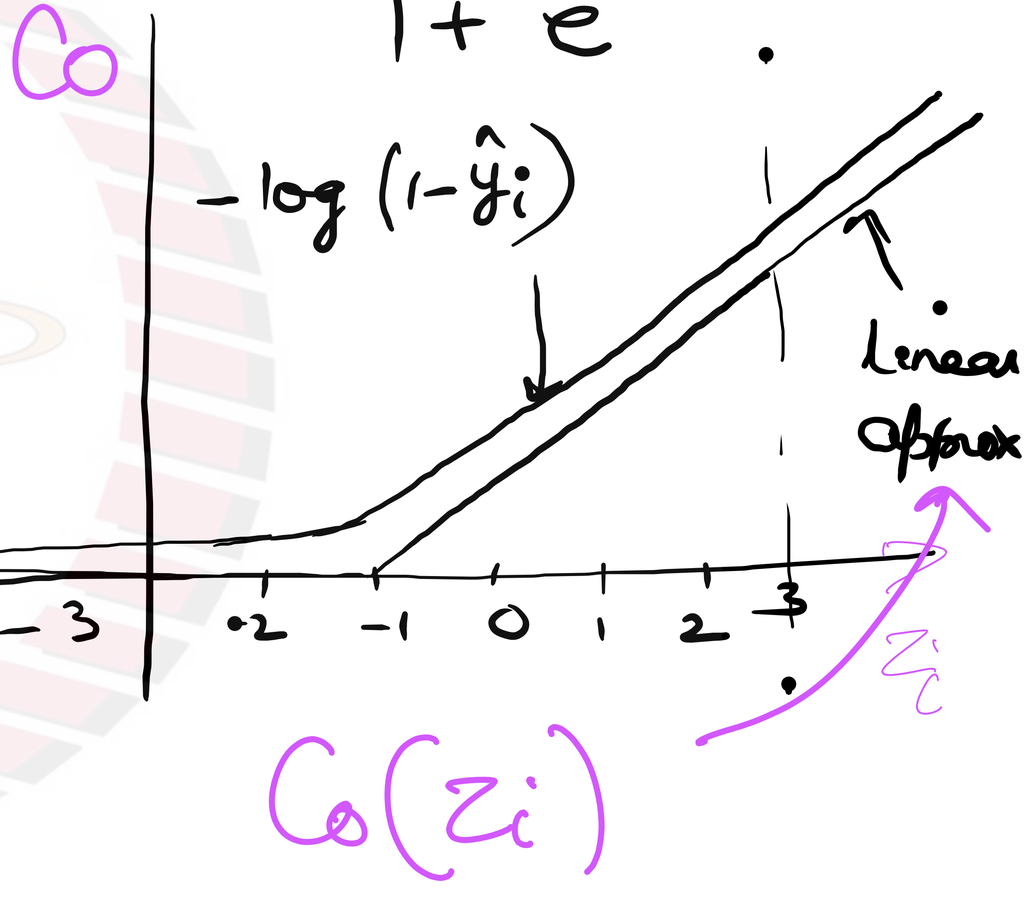
if $\hat{y}_i \to 0$ & $y_i = 1$

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if $y_i = 1$ $\quad -\log \hat{y}_i = -\log \frac{1}{1 + e^{-(x_i^T \theta + b)}}$

if $y_i = 0$ $\quad -\log(1 - \hat{y}_i) = -\log \frac{e^{-(x_i^T \theta + b)}}{1 + e^{-(x_i^T \theta + b)}}$





NPTEL

Linear SVM

Support Vector Machine

Training $\Rightarrow$ n no. of training samples

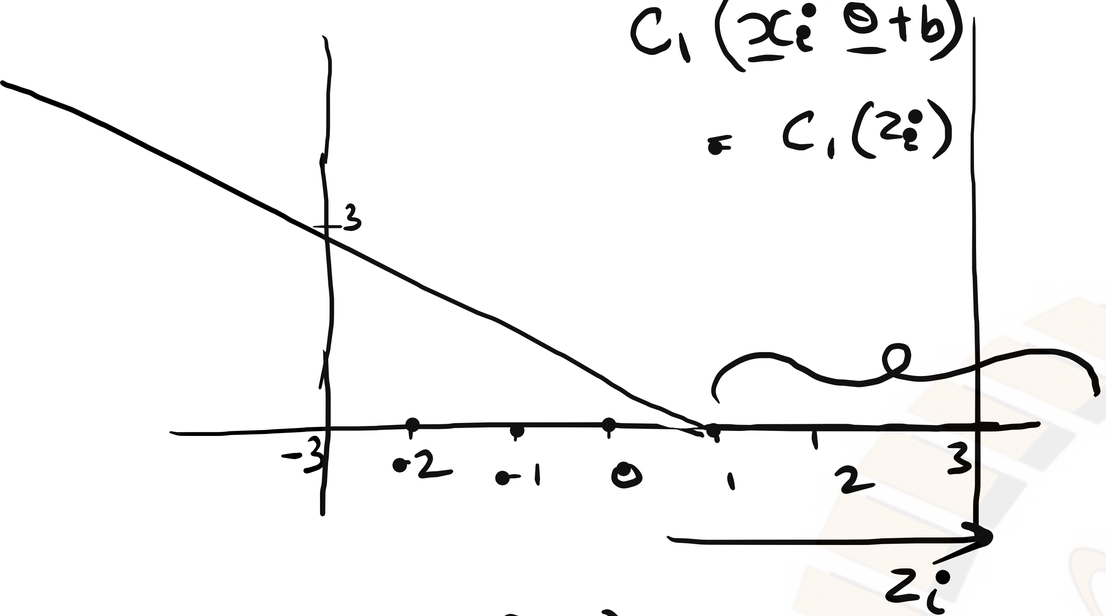
$$\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}_{p\times 1} \equiv \text{feature vector}$$

$$\mathcal{L}'(\underline{\theta}) = \eta \underbrace{\sum_{i=1}^{n} \left(y_i C_1 (\underline{x_i}^T \underline{\theta} + b) + (1-y_i) C_0 (\underline{x_i}^T \underline{\theta} + b) \right)}_{a(\underline{\theta})} + \underbrace{\frac{1}{2} \sum_{j=1}^{p} \theta_j^2}_{r(\underline{\theta}) \equiv \text{regularization term}}$$

(Ridge)

$$= \eta\, a(\underline{\theta}) + r(\underline{\theta})$$

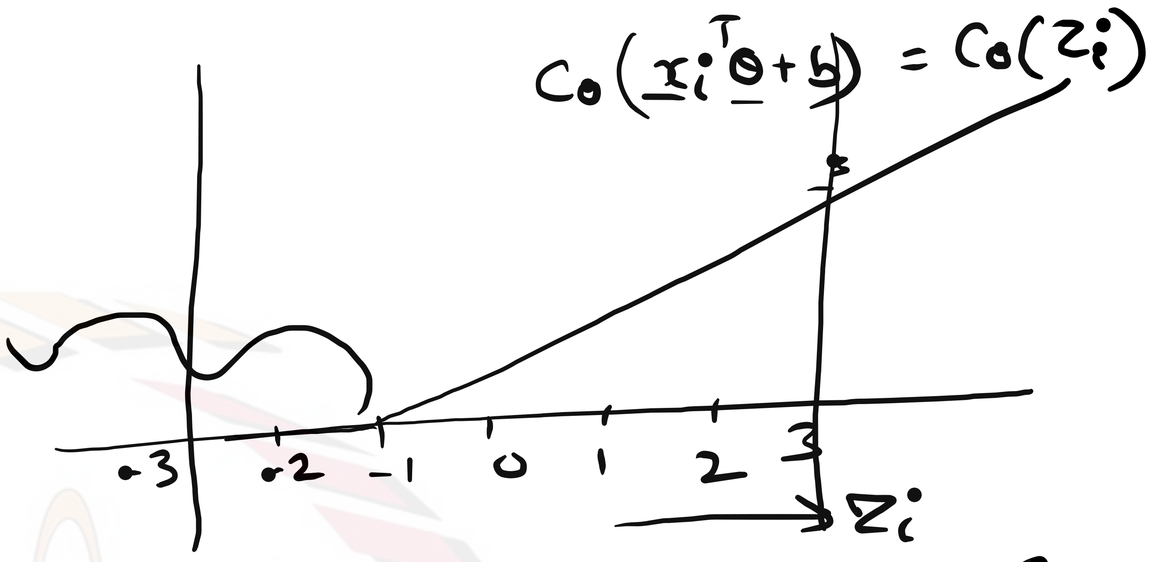
$$\underline{x_i}^T \underline{\theta} = \langle \underline{x_i}, \underline{\theta} \rangle = x_{i1}\theta_1 + x_{i2}\theta_2 + \cdots + x_{ip}\theta_p$$



$C_1(z_i) = 0$ if $z_i \geq 1$

if $y_i = 1$, we would like $z_i \geq 1$ (not satisfied) ≥ 0

if $y_i = 0$, $z_i \leq -1$ (not satisfied) ≤ 0



$C_0(z_i) = 0$ if $z_i \leq -1$

$\longrightarrow$ thus, we would like to increase the margin between class 0 & class 1.

$$\mathcal{L}'(\underline{\theta}) = \eta\, a(\underline{\theta}) + r(\underline{\theta})$$

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Case I : η is very large.

— we are paying more attention to $a(\underline{\theta})$

$\mathcal{L}'(\underline{\theta}) \longrightarrow$ low

$a(\underline{\theta}) \longrightarrow 0$

If $a(\underline{\theta}) = 0$ whenever $y_i = 1$ $\quad \underline{x}_i^T \underline{\theta} + b \geq 1$

& if whenever $y_i = 0$ $\quad \underline{x}_i^T \underline{\theta} + b \leq -1$

If this happens-

$$\min_{\underline{\theta}} \mathcal{L}'(\underline{\theta}) = \min_{\underline{\theta}} \frac{1}{2} \sum_{j=1}^{p} \theta_j^2$$

s.t. $\begin{cases} \underline{x}_i^T \underline{\theta} + b \geq 1 & \text{if } y_i = 1 \\ \underline{x}_i^T \underline{\theta} + b \leq -1 & \text{if } y_i = 0 \end{cases}$

if class 1: $y_i = 1$

if class 0: $y_i = -1$

s.t. $\boxed{y_i (\underline{x}_i^T \underline{\theta} + b) \geq 1}$

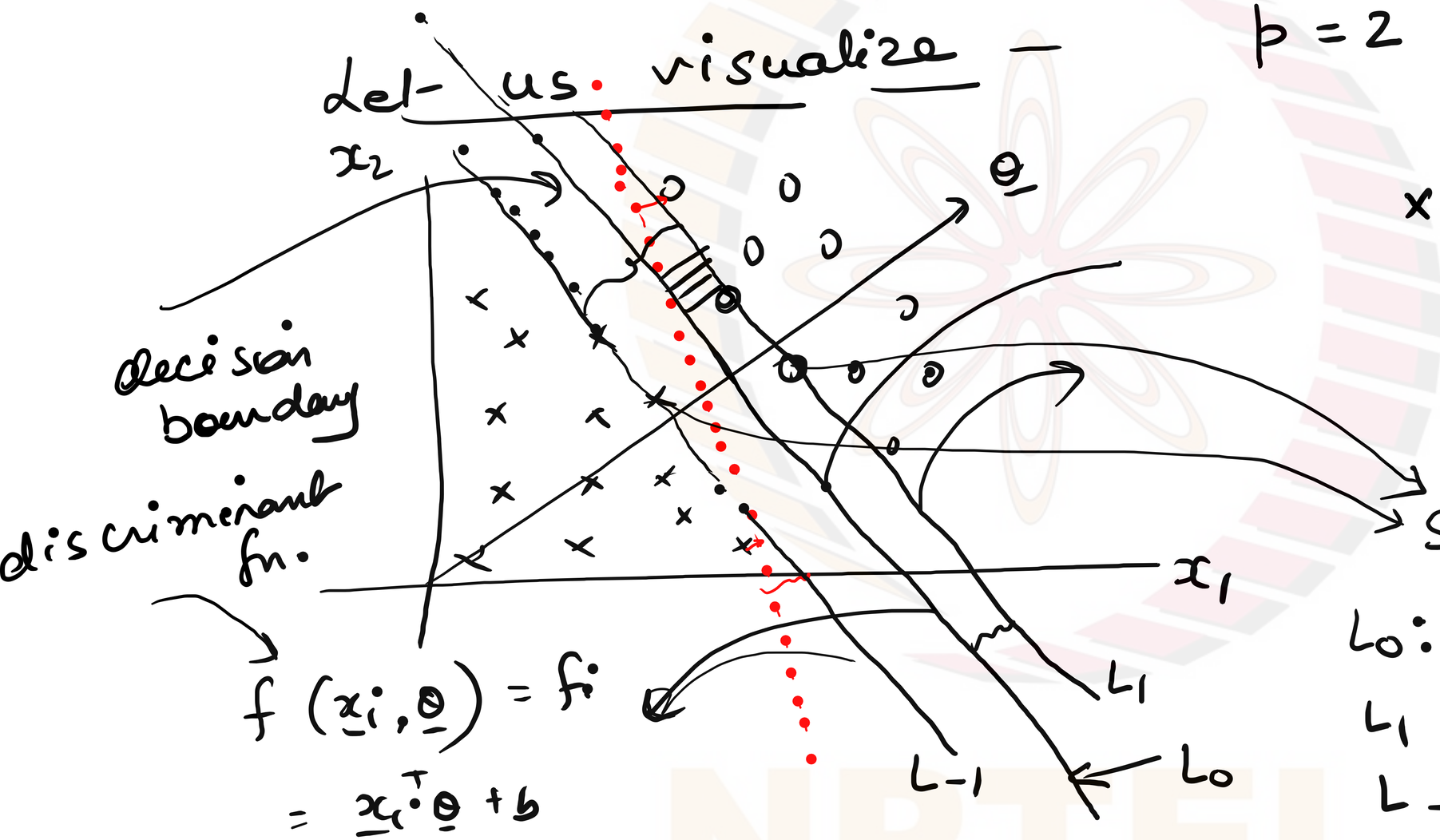
Hard margin classifier

Optimization problem for a hard-margin classifier –

$$\min_{\underline{\theta}} \ \frac{1}{2} \sum_{j=1}^{p} \theta_j^2 \qquad \text{s.t.} \quad y_i \left(\underline{x}_i^T \underline{\theta} + b \right) \geq 1$$

$\forall \ i = 1, 2 \ldots, n$

Let us visualize –



$f(\underline{x}_i, \underline{\theta}) = f_i$

$= \underline{x}_i^T \underline{\theta} + b$

$p = 2$ feature space = 2D

$x \equiv$ class -0, $y_i = -1$

$o \equiv$ class -1, $y_i = 1$

Support vectors

L_0: $\underline{x}_i^T \underline{\theta} + b = 0 \Rightarrow f_i = 0$

L_1 = $\underline{x}_i^T \underline{\theta} + b = 1 \Rightarrow f_i = 1$

L_{-1} = $\underline{x}_i^T \underline{\theta} + b = -1 \Rightarrow f_i = -1$

for a linearly separable classifier -

$$f(\underline{x}_i, \underline{\theta}) = \underline{x}_i^T \underline{\theta} + b$$

— In 2D feature space → straight line

→ $\underline{\theta}$ ⊥ decision boundary ⇒ parameter vector Or weight vector

→ if 3D feature space → decision boundary ⇓ plane

→ In general, P-Dimensional space it is called a hyperplane

linear

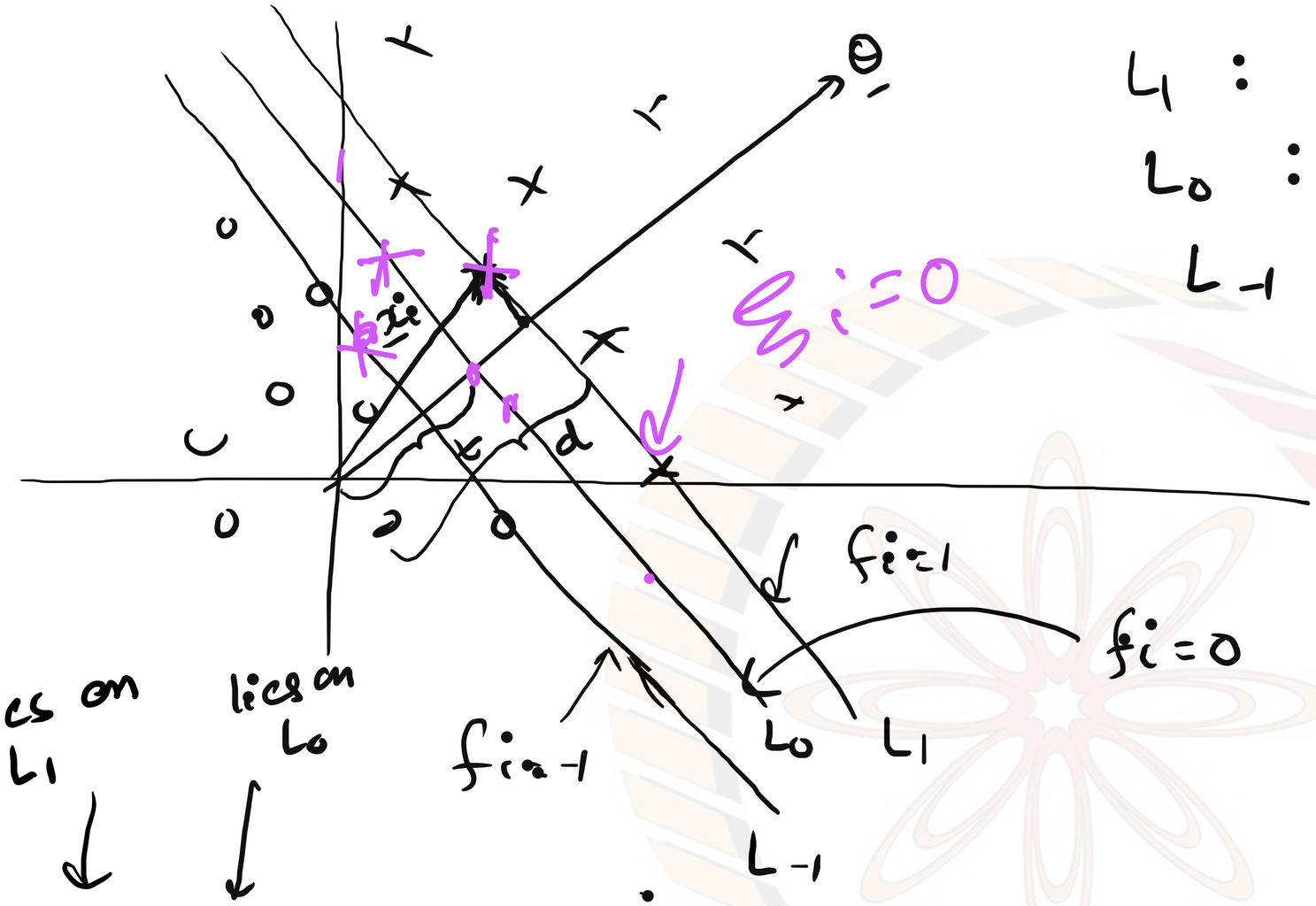
↑

↓

less than feature space dimensi

—

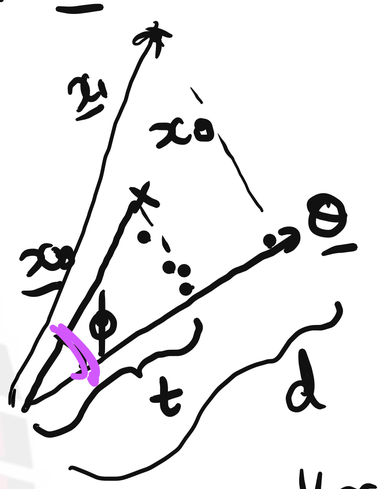
$$\max\left[0, \left(1 - y_i(x_i^T\theta + b)\right)\right] = \xi_i$$



$L_1 : \underline{x}_i^T \underline{\theta} + b = 1$ ↙ f_i

$L_0 : \underline{x}_i^T \underline{\theta} + b = 0$

$L_{-1} : \underline{x}_i^T \underline{\theta} + b = -1$



$\langle \underline{x}_0, \underline{\theta} \rangle = \|\underline{x}_0\| \|\underline{\theta}\| \cos\phi$

$\underline{x}_0^T \underline{\theta} = (\|\underline{x}_0\| \cos\phi) \|\underline{\theta}\|$

$P_{\underline{\theta}}(\underline{x}_0)$ t

$\underline{x}_0^T \underline{\theta} + b = 0 \Rightarrow \underline{x}_0^T \underline{\theta} = -b$

$\frac{\underline{x}_0^T \underline{\theta}}{\|\underline{\theta}\|} = t$

$t = \frac{\underline{x}_0^T \underline{\theta}}{\|\underline{\theta}\|} = \frac{-b}{\|\underline{\theta}\|}$

$d = \frac{\underline{x}_i^T \underline{\theta}}{\|\underline{\theta}\|}$

lies on L_1 — lies on L_0

$d - t =$ margin

For an pt. $\underline{x}_0$ which lies on L_0 —

$$\boxed{\underline{x_i}^T \underline{\theta} + b = 1} \leftarrow L_1$$

$$\Rightarrow \quad \underline{x_i}^T \underline{\theta} + b = 1$$

for any pt. lying on L_1 —

$$d = \frac{\underline{x_i}^T \underline{\theta}^*}{\|\underline{\theta}\|}$$

$$d - t = \frac{\underline{x_i}^T \underline{\theta}}{\|\underline{\theta}\|} - \left(\frac{-b}{\|\underline{\theta}\|}\right)$$

$$= \frac{\underline{x_i}^T \underline{\theta} + b}{\|\underline{\theta}\|} = \boxed{\frac{1}{\|\underline{\theta}\|}} \equiv \text{geometric margin}$$

↑ margin

distance between L_0 & L_1

SVM —

$$L'(\underline{\theta}) = \min_{\underline{\theta}} \left(r(\underline{\theta}) + \eta\, a(\underline{\theta}) \right)$$

Case-1 : η is very large —

$a(\underline{\theta}) = 0$

Hard margin classifier —

$$\min_{\underline{\theta}} \ \frac{1}{2} \|\underline{\theta}\|_2^2 \quad \text{s.t.} \quad a(\underline{\theta}) = 0$$

$$y_i(\underline{x}_i^T \underline{\theta} + b) \geq 1 \quad \forall i \ (i = 1, 2, \ldots n)$$

Class-0 $y_i = -1$

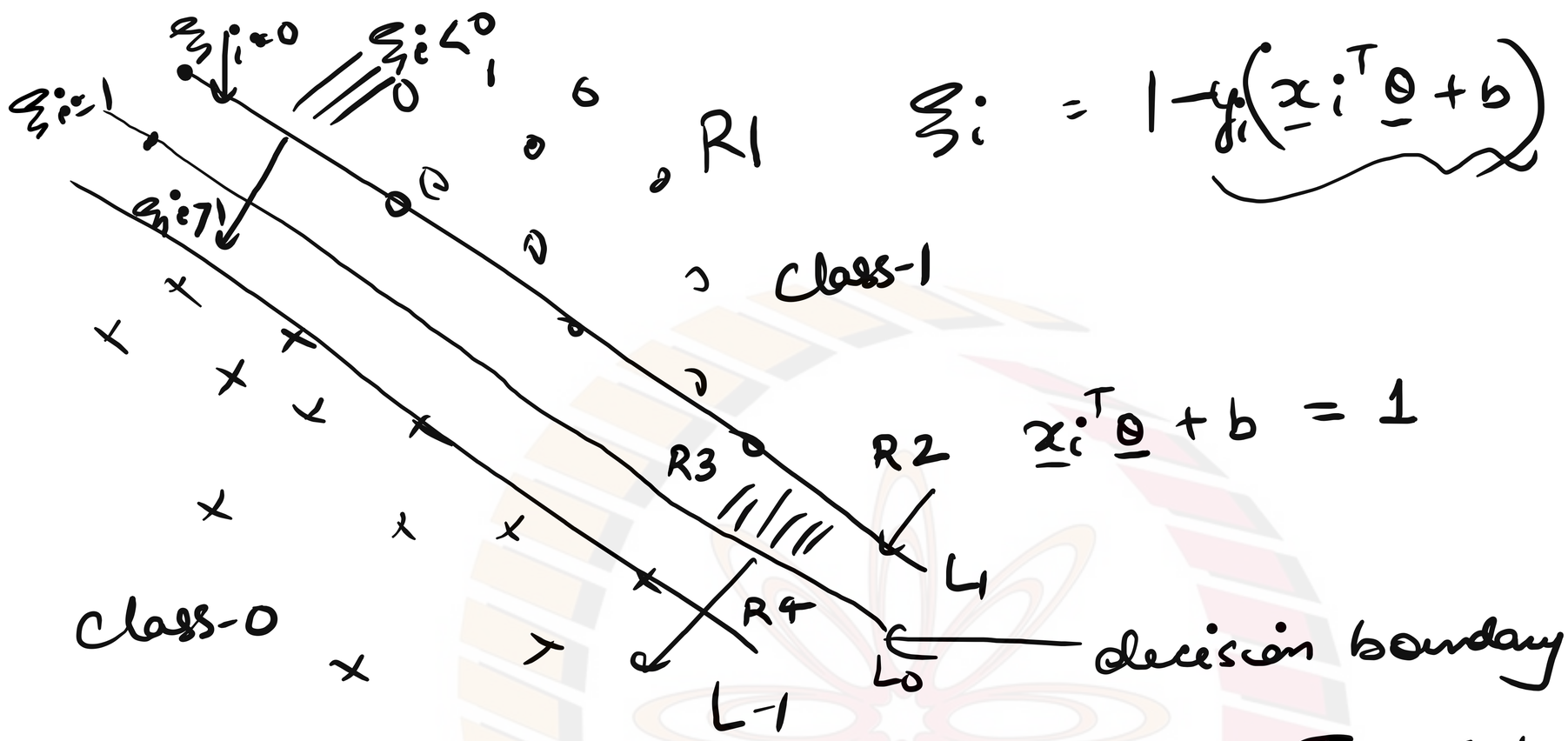
Class-1 $y_i = 1$

Case-2 : η is not very large.

$$L'(\underline{\theta}) = \min_{\underline{\theta}} \left(\frac{1}{2} \|\underline{\theta}\|_2^2 + \eta \sum_{i=1}^{n} \max\left(0, 1 - y_i(\underline{x}_i^T \underline{\theta} + b)\right) \right)$$

Slack variable $\xi_i = 1 - y_i(\underline{x}_i^T \underline{\theta} + b)$

Soft-margin classifier



$$\xi_i = 1 - y_i(x_i^T\theta + b)$$

between L_0 & L_1, for class-1, $0 < \xi_i < 1$

at L_0 $\xi_i = 1$

R1 region : $\xi_i < 0$ → correct classificati

R2 region: L_1 : $\xi_i = 0$ → " "

R3 " : $0 < \xi_i < 1$ → " " ← margin violation

R4 " $\xi_i > 1$ → mis classificati

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SVM primal formulation

$$\mathcal{L}'(\underline{\theta}) = \min_{\underline{\theta}} \frac{1}{2} \|\underline{\theta}\|_2^2 + \eta \sum_{i=1}^{n} \max(0, \xi_i)$$

ξ_i: slack variables

SVM dual formulation

Lagrangian multiplier —

$$\mathcal{L}'(\underline{\theta}, \underline{\eta}) = \min_{\underline{\theta}} \left(\frac{1}{2} \|\underline{\theta}\|_2^2 + \sum_{i=1}^{n} \eta_i \max(0, \xi_i) \right) \quad \eta_i \geq 0$$

$$= \min_{\underline{\theta}} \left(\frac{1}{2} \|\underline{\theta}\|_2^2 + \sum_{i=1}^{n} \eta_i \xi_i \right)$$

$$\underline{\eta} = \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_n \end{bmatrix}$$

$$\underline{\theta} = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_b \end{bmatrix}$$

$$\frac{\partial \mathcal{L}'(\underline{\theta}, \underline{\eta})}{\partial \underline{\theta}} = 0$$

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$$\rightarrow \alpha'(\underline{\theta}, \underline{\eta}) = \frac{1}{2}\underline{\theta}^T\underline{\theta} + \sum_{i=1}^{n} \eta_i \left\{1 - y_i\left(\underline{x_i}^T\underline{\theta} + b\right)\right\}$$

$$\frac{\partial \alpha'(\underline{\theta}, \underline{\eta})}{\partial \underline{\theta}} = \underline{\theta} - \sum_{i=1}^{n} \eta_i y_i \underline{x_i} = 0$$

$y_i \equiv$ label of ith sample

$\underline{x_i} \equiv$ feature vector

$$\frac{\partial \alpha'}{\partial b} = \boxed{\sum_{i=1}^{n} \eta_i y_i = 0} \Longrightarrow$$

$$\boxed{\underline{\theta}^* = \sum_{i=1}^{n} \eta_i y_i \underline{x_i}}$$

↑ representer theorem

θ^* can also be written as a linear combination of train data.

$$g(\underline{\eta}) = \inf_{\underline{\theta}} \alpha'(\underline{\theta}, \underline{\eta})$$

$$\boxed{\text{s.t. } g(\underline{\eta}) \leq p^*} \equiv \text{concave fn. in } \underline{\eta}$$

p^* ← where p^* is the acceptable or min cost.

(A) Hard margin classifier —

Primal $\min_{\underline{\theta}} \frac{1}{2} \underline{\theta}^T \underline{\theta}$ s.t. $y_i (\underline{x}_i^T \underline{\theta} + b) \geq 1 \quad \forall i = 1, 2 \ldots n$

Dual $\max_{\underline{\eta}} \min_{\underline{\theta}} \frac{1}{2} \left(\sum_{i=1}^{n} \eta_i y_i \underline{x}_i \right)^T \left(\sum_{j=1}^{n} \eta_j y_j \underline{x}_j \right)$ s.t. $y_i \left(\underline{x}_i^T \sum_{j=1}^{n} \eta_j y_j \underline{x}_j + b \right) \geq 1 \quad \forall i$

Dual $\Rightarrow$ $\max_{\underline{\eta}} \min_{\underline{\theta}} \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \eta_i y_i \eta_j y_j (\underline{x}_i^T \underline{x}_j)$

s.t. $\sum_{j=1}^{n} \eta_j y_i y_j \underline{x}_i^T \underline{x}_j + b \geq 1 \quad \forall i$

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(B) Soft Margin Classifier -

Primal $\min_{\underline{\theta}} \; \frac{1}{2} \underline{\theta}^T \underline{\theta} + \eta \sum_{i=1}^{n} \max(0, \xi_i)$

$$\underline{\eta} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \end{bmatrix}$$

Dual $\max_{\underline{\theta}} \; g(\underline{\eta})$

$$\mathcal{L}'(\underline{\theta}, \underline{\eta}) = \frac{1}{2} \underline{\theta}^T \underline{\theta} - \sum_{i=1}^{n} \eta_i \left[(\underline{x}_i^T \underline{\theta} + b) y_i - 1 \right]$$

$$g(\underline{\eta}) = \frac{1}{2} \left(\sum_{i=1}^{n} \eta_i y_i \underline{x}_i^T \right) \left(\sum_{j=1}^{n} \eta_j y_j \underline{x}_j \right) - \sum_{i=1}^{n} \eta_i y_i \underline{x}_i^T \left(\sum_{j=1}^{n} \eta_j y_j \underline{x}_j \right) - \sum_{i=1}^{n} \eta_i b y_i + \sum_{i=1}^{n} \eta_i$$

($\sum_{i=1}^{n} \eta_i b y_i = 0$)

But $\sum_{i=1}^{n} \eta_i y_i = 0$

$$\text{Dual } \max_{\underline{\eta}} g(\underline{\eta}) = \max_{\underline{\eta}} \left(\sum_{i=1}^{n} \eta_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \eta_i \eta_j y_i y_j (\underline{x}_i^T \underline{x}_j) \right)$$